Sentiment Analysis of Restaurant Reviews (Majhitar, Sikkim)

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1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to identify the emotional tone of text. This project analyzes restaurant reviews from Majhitar, Sikkim and classifies them into Positive, Neutral, and Negative categories.

2. Objectives

- Analyze real-world reviews
- Perform preprocessing
- Apply TF-IDF
- Train model
- Evaluate performance
- Deploy using Streamlit

3. Dataset

The dataset contains ~70 Google reviews with ratings (1–5). Ratings are mapped to sentiments:
4–5: Positive
3: Neutral
1–2: Negative

4. Data Preprocessing

Steps include lowercasing, removing special characters, tokenization, and stopword removal using NLTK. Limitations include no stemming/lemmatization and no negation handling.

5. Feature Engineering

TF-IDF was used to convert text into numerical features, capturing word importance.

6. Model

Logistic Regression was used due to its simplicity and efficiency for text classification.

7. Parameters

Test Size: 0.2
Random State: 42
Max Iterations: 200
Class Weight: Balanced

8. Evaluation

Accuracy achieved: ~46–48%. Model shows bias toward positive predictions due to class imbalance.

9. Deployment

Deployed using Streamlit: Input → Preprocessing → TF-IDF → Model → Prediction

10. Challenges

Small dataset, class imbalance, deployment issues, model loading errors.

11. Conclusion

An end-to-end NLP pipeline was successfully implemented. Performance is moderate due to dataset limitations.

12. Future Work

Increase dataset, apply lemmatization, use n-grams, try advanced models (SVM, LSTM, BERT), and deploy using FastAPI.

13. Example Predictions

Amazing food → Positive
It was okay → Neutral
Very bad experience → Negative

14. Summary

Pipeline: Data → Preprocessing → Feature Engineering → Model → Evaluation → Deployment